

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 3: TISSUES IN ACTION [SET B - ADVANCED HOTS]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH03/SET-B****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS (SET B - HIGHER ORDER THINKING SKILLS):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Advanced Conceptual and Numerical MCQs with single correct option.
3. **Section B (Q16 – Q20):** Critical Assertion-Reasoning based questions testing analytical clarity.
4. **Section C (Q21 – Q25):** Case-Study / Practical Experimental Data-interpretation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: ADVANCED & NUMERICAL MCQS (Q.1 TO Q.15)**[15 MARKS]**

Q.1 Which biomechanical feature enables the young green branches of a weeping willow tree to bend drastically in strong gales without snapping?

- (A) Lignified sclerenchyma fibres (B) Pectin-thickened corners of living collenchyma cells
(C) Gas-filled lacunae of aerenchyma (D) Dead suberized cork cells

Q.2 In a ringing (girdling) experiment, a ring of bark down to the vascular cambium is removed from the trunk of a woody tree. What is the immediate physiological consequence?

- (A) Water transport to the leaves stops immediately and the tree wilts within 1 hour (B) Downwards translocation of sugars is blocked, causing swelling of bark above the girdle while roots eventually starve
(C) Transpiration stops completely (D) Root hairs immediately burst

Q.3 Mature sieve tube elements lack a nucleus, ribosomes, and vacuoles. How do they maintain their metabolic activity over long periods?

- (A) Through independent photosynthesis (B) Via cytoplasmic plasmodesmatal connections with adjacent nucleated companion cells
(C) By absorbing enzymes from dead xylem vessels (D) They do not perform any living functions

Q.4 Stone cells (brachysclereids) responsible for the gritty texture when chewing the flesh of pear fruit are categorized under:

- (A) Collenchyma (B) Sclerenchyma
(C) Parenchyma (D) Chlorenchyma

Q.5 Which type of epithelial tissue lines the proximal convoluted tubules (PCT) of nephrons in the human kidney to maximize reabsorption?

- (A) Simple cuboidal epithelium with brush border microvilli (B) Stratified squamous keratinized epithelium
(C) Ciliated columnar epithelium (D) Sensory neuroepithelium

Q.6 Microscopic cross-sections of mammalian compact bone reveal concentric rings of lamellae surrounding central neurovascular channels termed:

- | | |
|-------------------------|---------------------|
| (A) Haversian canals | (B) Volkmann canals |
| (C) Chondrocyte lacunae | (D) Canaliculi only |
-

Q.7 Cartilage is slower to heal after sports injury compared to bone tissue primarily because:

- | | |
|---|---|
| (A) Cartilage matrix lacks calcium | (B) Cartilage is avascular (lacks direct blood vessels) and relies on slow diffusion through its matrix |
| (C) Chondrocytes divide much faster than osteocytes | (D) Cartilage is surrounded by periosteum |
-

Q.8 What specialized intercellular structures in cardiac muscle allow rapid ionic coupling and synchronized contractions of the heart chambers?

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|------------------------------|---|
| (A) Tight junctions only | (B) Intercalated discs with gap junctions |
| (C) Desmosomes without pores | (D) Neuromuscular motor endplates |
-

Q.9 Saltatory conduction of electrical nerve impulses in human motor neurons occurs because:

- | | |
|--|---|
| (A) The axon lacks a cell membrane | (B) The myelin sheath acts as an electrical insulator, forcing action potentials to jump between uninsulated Nodes of Ranvier |
| (C) Dendrites are longer than the axon | (D) Synapses contain chemical neurotransmitters |
-

Q.10 Which connective tissue cell type synthesizes histamine, heparin, and serotonin during allergic and inflammatory responses?

- | | |
|----------------------------|----------------|
| (A) Fibroblasts | (B) Mast cells |
| (C) Macrophage histiocytes | (D) Adipocytes |
-

Q.11 Intercalary meristem in monocot grasses is characteristically situated at:

- | | |
|--------------------------------------|---|
| (A) The extreme tip of the main root | (B) The base of internodes and leaf blades, facilitating rapid regrowth after grazing |
| (C) The bark cambium | (D) The floral petal margins |
-

Q.12 Which of the following elements of phloem tissue is dead at functional maturity?

- | | |
|-------------------------|---------------------------------|
| (A) Sieve tube elements | (B) Companion cells |
| (C) Phloem parenchyma | (D) Phloem fibres (bast fibres) |
-

Q.13 The lining of the human urinary bladder is composed of which specialized epithelial tissue capable of considerable distension without tearing?

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|--|--|
| (A) Simple squamous | (B) Transitional epithelium (urothelium) |
| (C) Pseudostratified ciliated columnar | (D) Stratified columnar |
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Q.14 When a sprinter runs a 100 m race, which muscle type provides rapid, explosive contractions but fatigues rapidly due to lactic acid build-up?

- | | |
|--------------------|---------------------------------|
| (A) Smooth muscle | (B) Striated skeletal muscle |
| (C) Cardiac muscle | (D) Visceral involuntary muscle |
-

Q.15 In desert xerophytic plants, water loss via cuticular transpiration is minimized by a thick waxy waterproof layer called:

- | | |
|-------------|------------|
| (A) Suberin | (B) Lignin |
| (C) Cutin | (D) Chitin |
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SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)**[5 MARKS]**

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): Tracheids and vessels in xylem are dead cells with lignified walls at maturity.

Reason (R): Dead hollow tubes without living cytoplasm reduce frictional resistance to the upward transpirational pull of water.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): Sprains are painful injuries caused by excessive stretching or tearing of ligaments.

Reason (R): Ligaments are tough fibrous bands that connect muscles to bones with zero elasticity.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): Cardiac muscle cells never exhibit fatigue under normal physiological conditions.

Reason (R): Cardiac myocytes have abundant mitochondria and a rich capillary blood supply enabling continuous aerobic respiration.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): Complex tissues consist of more than one type of cell working together as a functional unit.

Reason (R): Parenchyma, collenchyma, and sclerenchyma are examples of complex permanent tissues.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.20 Assertion (A): Cork cambium replaces epidermis as a woody tree grows older.

Reason (R): Cork cells are compactly arranged without intercellular spaces and their walls are suberized.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

SECTION C: EXPERIMENTAL DATA & CASE STUDY (Q.21 TO Q.25)**[5 MARKS]****READING PASSAGE & EXPERIMENTAL SETUP:**

Case Study: Microscopic Identification of Animal Tissues

A medical histology technician examines four unknown tissue biopsy slides labelled W, X, Y, and Z under high-power microscopy:

- **Slide W:** Long, unbranched, cylindrical syncytial fibres displaying distinct alternating light and dark transverse striations with peripheral nuclei.
- **Slide X:** Spindle-shaped cells with pointed tapering ends and a single central nucleus, showing no transverse striations.
- **Slide Y:** Solid mineralized matrix arranged in concentric osteons with spider-like osteocytes trapped inside lacunae.
- **Slide Z:** Clear jelly-like chondroitin matrix with cells grouped in pairs inside fluid spaces, found at the tip of the human nose.

Q.21 Tissue W is identified as:

- (A) Smooth muscle (B) Skeletal (striated) voluntary muscle
(C) Cardiac muscle (D) Dense regular connective tissue

Q.22 Where in the human body is Tissue X naturally located?

- (A) Attached to the femur bone (B) Walls of the stomach, intestine, and iris of the eye
(C) Heart ventricle wall (D) Biceps brachii

Q.23 Tissue Y is identified as:

- (A) Hyaline cartilage (B) Compact bone tissue
(C) Adipose tissue (D) Areolar tissue

Q.24 The chondrocytes trapped in pairs in Slide Z characterize which tissue?

- (A) Cartilage (B) Tendon
(C) Ligament (D) Blood

Q.25 Which tissue among W, X, Y, or Z is under voluntary somatic nervous control?

- (A) Tissue W only (B) Tissue X only
(C) Tissue W and X (D) Tissue Y and Z

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET B)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & IN-DEPTH EXPLANATIONS (SET B)

Q. No.	Correct	In-Depth Conceptual & Numerical Solution
Q.1	(B)	Collenchyma provides mechanical tensile strength combined with high elasticity and flexibility due to uneven pectin-cellulose wall thickenings.
Q.2	(B)	Removing bark removes phloem while leaving xylem intact. Water ascends normally, but downward transport of organic food to roots is severed, causing food accumulation above the girdle.
Q.3	(B)	Companion cells and sieve tube elements are sister cells derived from the same mother cell; companion cell nuclei and organelles metabolically sustain the enucleated sieve tubes.
Q.4	(B)	Sclereids are short, heavily lignified sclerenchyma cells with narrow lumens and branched pits, giving gritty texture to pears and sapotas.
Q.5	(A)	Cuboidal epithelium equipped with a microvilli brush border dramatically amplifies luminal surface area for reabsorption of glucose, ions, and water in renal PCT.
Q.6	(A)	Haversian canals contain blood vessels and nerve fibers running longitudinally through osteons, supplying nutrients to living osteocytes.
Q.7	(B)	Cartilage is an avascular tissue; nutrients must diffuse slowly through the chondroitin sulfate matrix from the surrounding perichondrium, slowing repair.
Q.8	(B)	Intercalated discs contain gap junctions (low-resistance electrical pathways) that allow action potentials to spread rapidly between cardiac myocytes.
Q.9	(B)	Myelin sheath acts as an insulator; depolarization can only occur at unmyelinated Nodes of Ranvier, accelerating impulse conduction speed up to 100 m/s.
Q.10	(B)	Mast cells in areolar tissue release histamine (vasodilator) and heparin (anticoagulant) during immune and allergic responses.
Q.11	(B)	Intercalary meristem is located at internode bases and leaf sheaths, enabling rapid regrowth of grass stems grazed by herbivores.
Q.12	(D)	Phloem fibres (bast fibres, such as commercial jute, flax, and hemp) are dead sclerenchymatous support elements; all other phloem components are living.
Q.13	(B)	Transitional epithelium can stretch and flatten when the bladder fills with urine, accommodating volume changes without tearing.
Q.14	(B)	Skeletal muscles (striated, voluntary) contract with high velocity and power, but anaerobic glycolysis under strenuous exertion produces lactic acid fatigue.
Q.15	(C)	Cutin is a waxy, hydrophobic polyester covering the epidermal layer of aerial plant surfaces, preventing excessive cuticular water loss.
Q.16	(A)	Hollow lumens devoid of protoplasmic obstructions allow unbroken capillary columns of sap to ascend rapidly under negative transpirational tension.
Q.17	(C)	Assertion is true. Reason is false: Ligaments connect bone to bone (tendons connect muscle to bone) and contain elastic fibers that permit limited joint movement.
Q.18	(A)	High mitochondrial volume fraction (≈40%) and continuous coronary perfusion ensure cardiac muscle relies on aerobic metabolism without accumulating lactic acid.
Q.19	(C)	Assertion is true. Reason is false because parenchyma, collenchyma, and sclerenchyma are simple permanent tissues (made of only one cell type). Xylem and phloem are complex tissues.
Q.20	(B)	Both statements are correct biological facts, but the reason explains why cork is an effective protective barrier, not the developmental trigger for cambial initiation.
Q.21	(B)	Multinucleated cylindrical fibers with peripheral nuclei and alternating A-bands and I-bands are characteristic of skeletal muscle.

Q. No.	Correct	In-Depth Conceptual & Numerical Solution
Q.22	(B)	Tissue X is smooth muscle (spindle-shaped, uninucleated, non-striated), located in involuntary visceral organs like the digestive tract.
Q.23	(B)	Concentric lamellae, osteons, and osteocytes embedded in hard mineralized calcium matrix define compact bone tissue.
Q.24	(A)	Chondrocytes residing in lacunae within an elastic chondroitin sulfate matrix constitute cartilage (e.g., nose tip, external ear pinna).
Q.25	(A)	Only skeletal muscle (Tissue W) can be consciously contracted by voluntary motor signals.